

LAB EXPERIMENT RECORD

SET 10: SURVEY RESPONSES ANALYSIS

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Aim

To analyze survey response data using a grouped bar chart, a stacked bar chart and a survey table in R, and to build an interactive dashboard using Tableau. Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

PROGRAM 1: Grouped Bar Chart of Question 1 Answers

Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

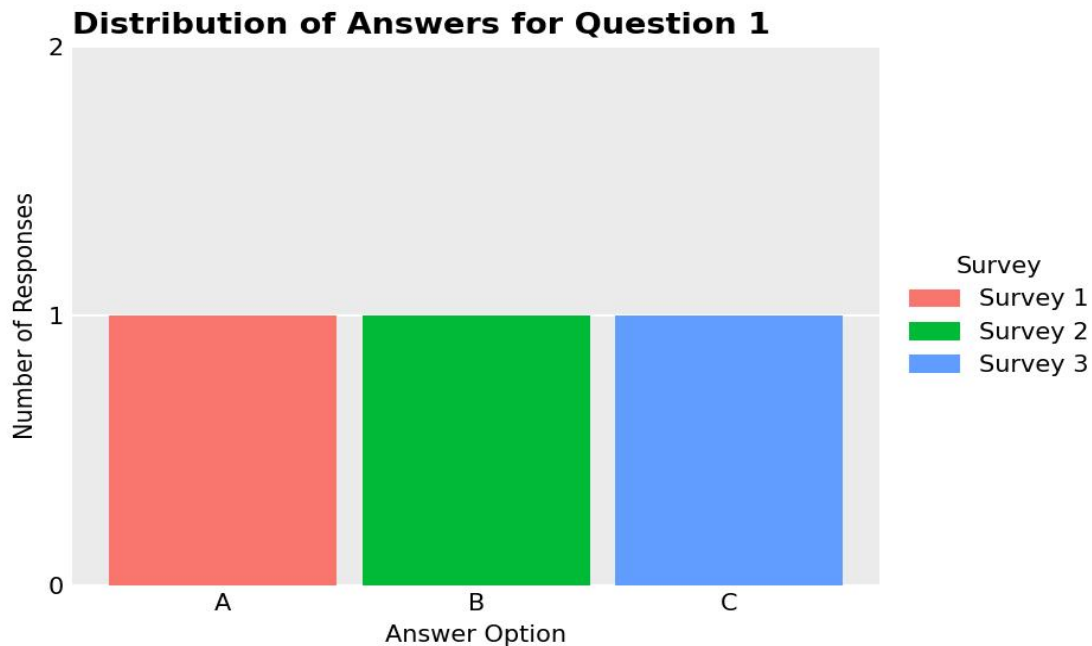
```
# Dataset
survey <- data.frame(
  Survey_ID = 1:3,
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

library(ggplot2)

# Answers given for Question 1, by survey
q1 <- data.frame(
  Answer = survey$Question_1,
  Survey = paste("Survey", survey$Survey_ID),
  Responses = 1
)

# Grouped Bar Chart
ggplot(q1, aes(x = Answer, y = Responses, fill = Survey)) +
  geom_col(position = "dodge") +
  scale_y_continuous(breaks = 0:2, limits = c(0, 2)) +
  labs(title = "Distribution of Answers for Question 1",
       x = "Answer Option",
       y = "Number of Responses",
       fill = "Survey")
```

OUTPUT:



Output 1: Distribution of Answers for Question 1 (Grouped Bar Chart)

Observation: Question 1 received three different answers, A, B and C, and each was chosen exactly once, by Survey 1, Survey 2 and Survey 3 respectively. All three bars are the same height, so no answer was more popular, and nobody chose option D.

PROGRAM 2: Stacked Bar Chart of Overall Responses

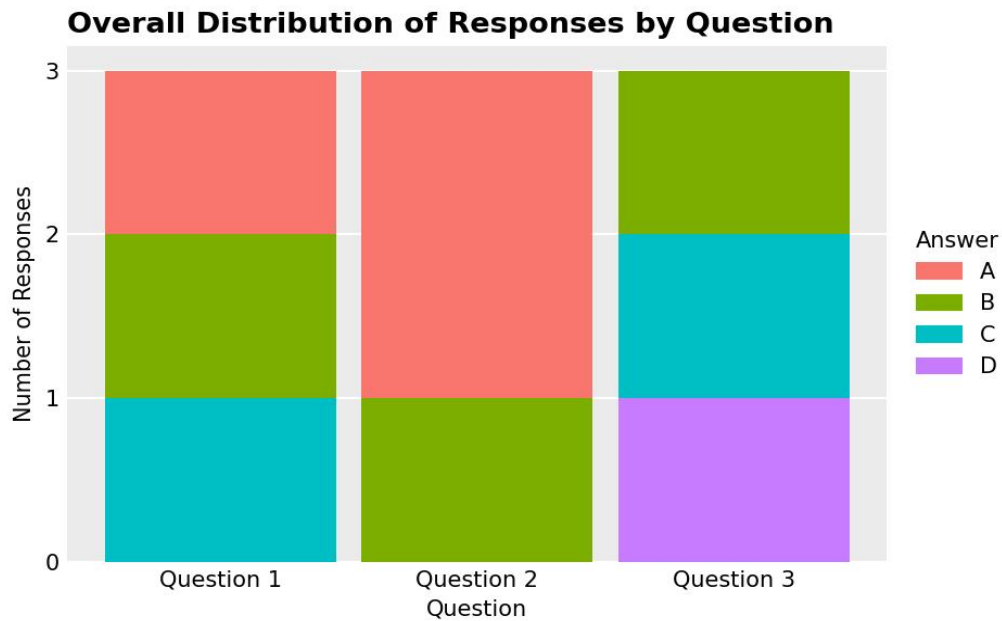
Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

```
library(ggplot2)

# Dataset in long format (one row per response)
long_survey <- data.frame(
  Question = rep(c("Question 1", "Question 2", "Question 3"), each = 3),
  Answer    = c("A", "B", "C",
                "B", "A", "A",
                "C", "D", "B")
)

# Stacked Bar Chart
ggplot(long_survey, aes(x = Question, fill = Answer)) +
  geom_bar() +
  scale_y_continuous(breaks = 0:3) +
  labs(title = "Overall Distribution of Responses by Question",
       x = "Question",
       y = "Number of Responses",
       fill = "Answer")
```

OUTPUT:



Output 2: Overall Distribution of Responses (Stacked Bar Chart)

Observation: Each question has 3 responses, giving 9 responses in total. Answers A and B are the most common, with 3 responses each (about 33% each), followed by C with 2 (about 22%) and D with 1 (about 11%). Question 2 is the least varied, with answer A chosen by two of the three respondents, while Question 3 has three different answers (B, C and D). Option D appears only in Question 3.

PROGRAM 3: Table of Survey Responses

Build a table to show the survey response data for each survey. Label the table elements.

```
# Dataset
survey <- data.frame(
  Survey_ID = 1:3,
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

# Display the survey table
print(survey)

# Count of each answer for every question
freq <- as.data.frame(matrix(table(long_survey$Answer, long_survey$Question)))
freq$Total <- rowSums(freq)
print(freq)

# Total responses for each question
colSums(freq)
```

Console Output

```

Survey_ID Question_1 Question_2 Question_3
1          1          A          B          C
2          2          B          A          D
3          3          C          A          B
  Question 1 Question 2 Question 3 Total
A          1          2          0      3
B          1          1          1      3
C          1          0          1      2
D          0          0          1      1
Question 1 Question 2 Question 3      Total
          3          3          3          9

```

Output 3a: Survey Responses Table (R Console)

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Output 3b: Survey Response Table

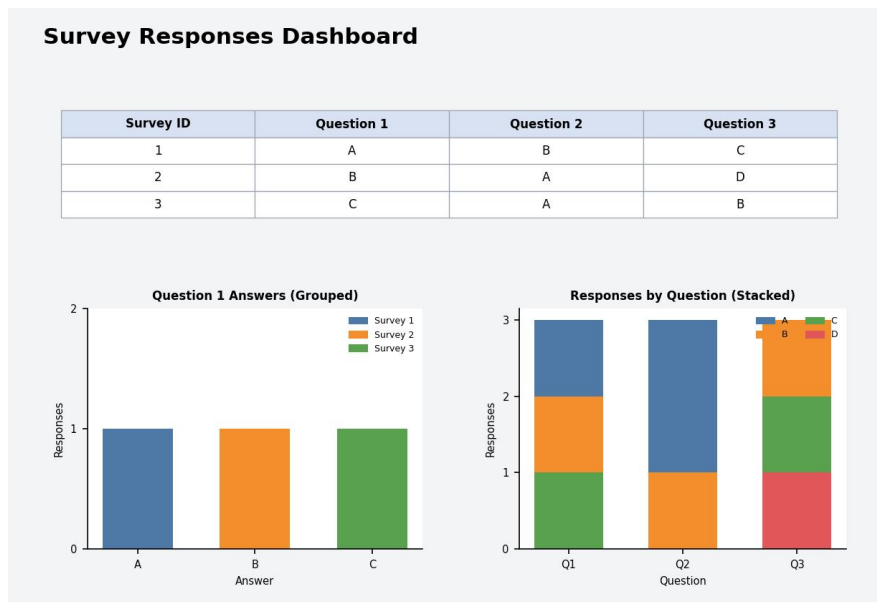
Observation: The table gives the exact answers of each survey to every question. Survey 1 answered A, B, C, Survey 2 answered B, A, D and Survey 3 answered C, A, B. The answer counts in the console output show that answers A and B were each given 3 times, C twice and D once, out of 9 responses in total.

PROGRAM 4: Interactive Survey Responses Dashboard in Tableau

Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart and the table for interactive exploration of survey responses.

Steps in Tableau

1. Save the data as an Excel or CSV file with the columns Survey ID, Question 1, Question 2 and Question 3, and connect to it in Tableau (Connect > Microsoft Excel / Text file). Select the Question 1, Question 2 and Question 3 columns, right-click and choose Pivot, then rename the new fields to "Question" and "Answer".
2. Sheet 1 (Grouped Bar Chart): Drag Answer to Columns and Number of Records (CNT) to Rows. Drag Survey ID to Color, add a filter on Question and select Question 1. Add the count to Label and rename the sheet "Question 1 Answers".
3. Sheet 2 (Stacked Bar Chart): Drag Question to Columns and Number of Records (CNT) to Rows. Drag Answer to Color so that the bars stack, add the count to Label, and rename the sheet "Responses by Question".
4. Sheet 3 (Table): Drag Survey ID to Rows, Question to Columns and Answer to Text. Rename the sheet "Survey Table".
5. Create a new Dashboard (Dashboard > New Dashboard), set the size to Automatic, and drag the three sheets onto the dashboard area.
6. Add interactivity: select each chart and click Use as Filter, so clicking an answer or a question in one view filters the others. Add Answer and Survey ID filters (Show Filter > Apply to Worksheets > All Using This Data Source).
7. Add the title "Survey Responses Dashboard" and test the filters in Presentation Mode.



Output 4: Interactive Survey Responses Dashboard (Tableau)

Observation: The dashboard shows the Question 1 answers, the overall distribution of responses and the survey table together. Clicking an answer or a question in one view filters the other views.

Procedure

1. Define the survey responses (Survey ID and the answers to three questions) in R as a data frame.
2. Use `ggplot2` with `geom_col(position = "dodge")` to create the grouped bar chart of answers to Question 1.
3. Arrange the data in long format and use `ggplot2` with `geom_bar()` to draw the stacked bar chart for all three questions.
4. Display the survey table with `print()`, count each answer for every question with `table()`, and total the responses with `rowSums()` and `colSums()`.
5. Load the survey file into Tableau, pivot the question columns, and build the grouped bar chart, stacked bar chart and table worksheets.
6. Combine the worksheets in a Tableau dashboard and add filters and actions for interactivity.

RESULT:

The survey response data was successfully analyzed using a grouped bar chart, a stacked bar chart and a survey table in R. An interactive Tableau dashboard was also created, combining all three views so that users can explore the survey responses by question and by answer.

Key Functions and Tools Used

Function / Tool	Purpose
<code>data.frame()</code>	Creates the survey dataset and long-format data.
<code>ggplot()</code> / <code>geom_col()</code>	Creates the grouped bar chart (<code>position = "dodge"</code>).
<code>ggplot()</code> / <code>geom_bar()</code>	Creates the stacked bar chart.
<code>print()</code> / <code>table()</code> / <code>colSums()</code>	Displays the survey table and counts the answers.
Tableau Pivot	Converts the question columns into Question and Answer fields.

Function / Tool	Purpose
Tableau Dashboard	Combines worksheets with filters and actions.